NanoROS

A Dual-Trigger Nanoparticle System for Precision Mitochondrial ROS Scavenging in Doxorubicin-Induced Cardiotoxicity

Pre-Submission Research Specification | Collaboration Outreach Document

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Abstract

Doxorubicin (dox) cardiotoxicity affects up to 40% of patients receiving standard anthracycline chemotherapy and remains without a precision pharmacological solution. Dexrazoxane reduces cardiac damage by approximately 40% through iron chelation but carries secondary leukemia risk; elamipretide (Forzinity, FDA approved September 2025) validates cardiolipin (CL)-binding as a mitochondrial therapeutic mechanism but fails in acquired cardiac disease due to absence of cardiac selectivity, no ROS-scavenging payload, and constitutive delivery without disease-state gating. This specification describes **NanoROS**: a four-layer programmable nanoparticle engineered to address all three mechanistic gaps simultaneously.

The design exploits dox-induced upregulation of ICAM-1 on cardiac endothelium (3-5-fold, Hayashi 2004) to create a self-amplifying cardiac accumulation mechanism. A dual-polymer AND gate -- dioxaborolane sensing sustained H_2O_2 elevation (Arm A, k_2 = 600 $M^{-}s^{-}$ at matrix pH 8.0) combined with O-alkylhydroxylamine oxime sensing 4-hydroxynonenal (4-HNE, Arm B, k_2 = 300-1,000 $M^{-}s^{-}$ depending on branch outcome) -- gates payload release against the co-occurrence of two independent pathological signals. The payload, MnTE-2-PyP, intercepts superoxide upstream (kcat/Km = 7×10^8 $M^{-}s^{-}$) with ceiling-level protection (~87%) when released before the ODE-defined damage deadline at t = 123 minutes.

Five pre-specified experimental gates determine which of three defined design branches is implemented. The document presents the complete contingency design for each gate outcome, ensuring no experimental result renders the program undefined.

Contents

1. Introduction and Rationale

1.1 The Clinical Problem

Doxorubicin is among the most effective cytotoxic agents in oncology. It anchors regimens for breast cancer, lymphoma, sarcoma, and acute leukemia. It is also irreversibly cardiotoxic. Up to 30-40% of patients receiving standard multi-cycle anthracycline chemotherapy develop measurable subclinical cardiac injury; 5-10% develop symptomatic cardiomyopathy. At cumulative doses 300 mg/m² or above, the incidence of clinical heart failure within a decade of treatment approaches 26% (Swain et al. 2003). In breast cancer survivors -- the population for whom life expectancy after treatment is longest -- cardiac disease has become the leading cause of non-cancer mortality (Armenian et al. 2019).

The mechanism is well characterised. Doxorubicin undergoes redox cycling at mitochondrial Complexes I and III, generating a sustained burst of superoxide ($O_2^{\cdot -}$) that SOD converts to hydrogen peroxide (H_2O_2). At the peak H_2O_2 concentrations predicted by our validated kinetic model (6.89 nM at t = 587 min post-dose), oxidative attack on mitochondrial cardiolipin (CL) produces CL hydroperoxides (CL-OOH). CL-OOH decomposition generates diffusible lipid aldehydes including 4-hydroxynonenal (4-HNE) and malondialdehyde (MDA). Cumulative damage to the electron transport chain, mitochondrial DNA, and contractile proteins produces irreversible cardiomyocyte loss.

The endogenous defence response is insufficient. Nrf2 -- the master regulator of antioxidant gene expression -- does not activate to 2-fold above baseline until t = 160 minutes post-dose in our model, well after the ODE-defined critical damage threshold (D = 50 AU at t = 123 minutes). GSH depletion under standard dox exposure is only 0.8% -- insufficient to serve as a trigger signal and insufficient to buffer the ROS burst.

1.2 The Inadequacy of Current Solutions

Dexrazoxane (Zinecard) is the only FDA-approved cardioprotectant for anthracycline therapy. It acts by chelating iron and preventing iron-mediated ROS amplification via Fenton chemistry. Clinical data demonstrate approximately 40% reduction in cardiotoxicity incidence (Swain 1997). Dexrazoxane is not adequate for two reasons. First, it does not address the direct mitochondrial ROS burden -- the 40% residual cardiotoxicity represents the fraction driven by Complex I/III superoxide generation independent of iron cycling. Second, it carries secondary risk: it cannot be co-administered with concurrent trastuzumab, and the DCOG AIEOP label warns of secondary acute myeloid leukaemia risk in paediatric populations.

Systemic antioxidants (vitamins C, E, N-acetylcysteine) fail because ROS is not uniformly pathological. Superoxide is required for neutrophil oxidative burst, hydrogen peroxide mediates proliferative signalling, and mitochondrial ROS is essential for hypoxic vasoconstriction. Systemic suppression of ROS produces off-target effects that partially or fully offset cardiac benefit.

MitoQ established proof-of-concept that mitochondrial-targeted antioxidant delivery could provide cardiac benefit. However, MitoQ has no viable therapeutic window at any mitochondrial membrane potential. At normal $\Delta\Psi m$ (150 mV), the 275-fold TPP^+ accumulation ratio means that a therapeutic plasma concentration of 100 nM produces an intramitochondrial concentration of 27.5 µM -- 27-fold above the ~1 µM pro-oxidant ceiling at which MitoQ ubisemiquinone cycling begins generating superoxide (Doughan & Dikalov, Antioxid Redox Signal 2007). MitoQ cannot be dosed to be simultaneously safe and effective across a heterogeneous population of mitochondria with varying $\Delta\Psi m$ -- a constraint that does not apply to MnTE-2-PyP, which has no semiquinone cycling side reaction and no established pro-oxidant ceiling at therapeutic concentrations.

1.3 Elamipretide and the Cardiolipin Opportunity

The cardiolipin-binding mechanism exploited for mitochondrial docking in this design was granted FDA accelerated approval on September 19, 2025, in the form of elamipretide (Forzinity, Stealth BioTherapeutics) -- the first approved therapy for Barth syndrome, a genetic mitochondrial cardiolipin disorder. This approval formally validates the biological premise of CL-targeting as a mitochondrial therapeutic mechanism.

The mechanistic basis for free elamipretide's failure in acquired cardiac disease identifies three specific gaps that the NanoROS architecture addresses. First, systemic administration provides no cardiac selectivity. Second, elamipretide's CL-stabilisation mechanism does not scavenge the superoxide that is peroxidising the cardiolipin it stabilises. Third, constitutive elamipretide activity stabilises CL in all tissues, including immune cells where ROS production is necessary. NanoROS resolves all three gaps: CRPPR + ICAM-1 targeting creates cardiac selectivity; MnTE-2-PyP intercepts $O_2^{\cdot -}$ at source; and the dual-trigger AND gate gates payload release against the disease state.

2. Novelty Claims

Five novelty claims are advanced. Each is defined with sufficient specificity to be falsified, and each rests on a body of prior work that validates its components while leaving the integration itself unpublished.

Claim 1: ODE-Model-Constrained Trigger Specification

No prior nanoparticle design for ROS-mediated cardioprotection has been parameterised against a validated mechanistic model of the damage it aims to prevent. The ROS-Cardio model, validated against Ludke et al. 2017 adult rat cardiomyocyte data, provides peak H_2O_2 (6.89 nM, t = 587 min), peak $O_2^{\cdot -}$ (2.51 nM), the damage deadline (D = 50 AU, t = 123 min), and the Nrf2 activation lag (2-fold at t = 160 min). These numbers determine dioxaborolane Arm A threshold, AND gate firing window, required k_scavenge, and minimum acceptable encapsulation efficiency. The quantitative constraint is the novelty.

Claim 2: Orthogonal Dual-Signal AND Gate

Prior ROS-sensitive nanoparticles use single-trigger designs. The NanoROS AND gate requires co-occurrence of sustained H_2O_2 elevation (dioxaborolane, temporal integrator, GSH-inert, acid-stable) and 4-HNE accumulation (O-alkylhydroxylamine oxime, lipid peroxidation marker, GSH-inert, acid-stable). The two chemistries are mechanistically orthogonal: boronate esters react with H_2O_2 but not aldehydes; hydroxylamines react with aldehydes but not H_2O_2. Specificity table modelling across eight biological scenarios shows zero false positives for Branches A and B.

Claim 3: Cardiolipin Coherence -- Single Lipid, Three Functions

SS-31 targets intact CL. CL peroxidation to CL-OOH weakens SS-31 binding affinity. CL-OOH beta-scission produces 4-HNE, activating AND gate Arm B. Three distinct functions -- mitochondrial docking, Arm B trigger generation, and NP detachment -- are mechanistically coupled through a single lipid's oxidation state. The therapeutic index is self-amplifying: as CL peroxidation increases with disease severity, the NP docks more specifically to oxidatively stressed mitochondria while simultaneously becoming more trigger-responsive.

Claim 4: ICAM-1 / 4-HNE Self-Amplifying Therapeutic Index

Doxorubicin-induced cardiac stress upregulates ICAM-1 on cardiac endothelium 3-5-fold (Hayashi et al. 2004) and simultaneously drives CL-OOH to 4-HNE accumulation in cardiomyocyte mitochondria. Both amplification mechanisms are driven by the same upstream cause. The NP accumulates more efficiently in hearts where ICAM-1 is higher (more disease), and its AND gate is more readily triggered where 4-HNE is higher (more disease). NanoROS becomes more cardioselective as disease severity increases.

Claim 5: 4-HNE Oxime Trigger as Novel NP Release Chemistry

To our knowledge, 4-HNE has not been used as a nanoparticle trigger molecule in any published nanomedicine design. All prior lipid-peroxidation-sensing NP designs have attempted to sense CL-OOH directly, encountering three unsolved problems: membrane-contact requirement, GSH competition, and acid instability. Sensing the diffusible downstream product 4-HNE via O-alkylhydroxylamine oxime ligation resolves all three: 4-HNE is freely diffusible, oxime ligation is GSH-inert, and oxime bonds are acid-stable ($t_{1/2} >> 24$ hr at pH 5.5, Kalia & Raines 2008).

3. Computational Foundation: The ROS-Cardio ODE Model

3.1 Model Structure

The ROS-Cardio model tracks eight state variables in a dox-exposed cardiomyocyte: [Dox], [$O_2^{\cdot -}$], [H_2O_2], Keap1 oxidation state, Nrf2 nuclear concentration, [GSH], cumulative irreversible damage D, and -- in the extended module -- CL-OOH and free 4-HNE. Core ODEs:

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d[Dox]/dt = k_in - k_cl*[Dox]
d[O2-]/dt = k_ros*[Dox] - k_sod*[O2-]
d[H2O2]/dt = k_sod*[O2-] - k_gsh*f(GSH)*[H2O2]
dD/dt = k_damage*[H2O2] - k_repair*D
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The Nrf2 module captures the feedback loop between oxidative stress, Keap1 oxidation, Nrf2 nuclear translocation, and GSH synthesis induction. The extended module adds CL peroxidation kinetics and 4-HNE production/clearance (k_HNE_clear = 3.0 min-1, derived from GSH Michael addition rate k2 = 10^4 $M^- s^-$ at [GSH] = 5,000 nM).

3.2 Validation

The model is fitted to Ludke et al. 2017 -- adult rat cardiomyocyte H_2O_2 dynamics under dox exposure. H_2O_2 trajectory is normalised to the published t = 1 hr baseline (120% of control) and the model tracks experimental data across 24 hours. The Nrf2 activation timing (2-fold at t = 160 min) matches published GSH synthesis induction kinetics in dox-treated cells.

3.3 Key Design Constraints

Parameter	Value	Design Use
Dox steady state	25 nM	Target cell context
Peak O2-	2.51 nM at t = 547 min	SOD mimic sizing
Peak H2O2	6.89 nM at t = 587 min	Trigger range upper bound
H2O2 at 10% peak	0.69 nM at t = 6 min	Trigger range lower bound
Nrf2 2x activation	t = 160 min	Endogenous defence too slow
GSH depletion	0.8% at standard dose	NOT a viable trigger signal
Damage deadline (D = 50 AU)	t = 123 min	NP must act before this
[CL-OOH]_ss (calibrated)	~5 nmol/mg	4-HNE production anchor
[4-HNE]_ss (estimated)	0.034-0.46 uM	Arm B trigger range
k_scavenge required	>= 5 min-1	Payload potency floor

4. Nanoparticle Architecture

NanoROS is an 80-120 nm diameter polymer-lipid hybrid nanoparticle with four concentric functional layers. Each layer has an independent mechanistic rationale; all four must function simultaneously and compatibly.

4.1 Layer 1 -- Systemic Delivery and Cardiac Accumulation

- **CRPPR targeting peptide** (Cys-Arg-Pro-Pro-Arg) at PEG chain termini provides cardiac endothelium homing. Dvir et al. 2011 demonstrated 4.2% cardiac accumulation in mice. Human-realistic estimate: 0.3-1.0% given species differences in cardiac output and endothelial biology.
- **Anti-ICAM-1 targeting** is the second and more strategically important accumulation mechanism. Doxorubicin upregulates ICAM-1 on cardiac endothelium 3-5-fold within hours of exposure (Hayashi et al. 2004). Anti-ICAM-1 NPs show 4-fold cardiac accumulation improvement over untargeted controls (Bhattacharya et al. 2012). More dox stress = more ICAM-1 = more NP accumulation.
- **CD47 mimetic** (Rodriguez et al. Science 2013) decorates the NP surface to inhibit macrophage phagocytosis, extending circulation half-life.
- **PEG corona** (2 kDa, hydrazone acid-labile linker, ~150 chains/NP) provides immune evasion and ~3-6 hr circulation half-life. The hydrazone linker hydrolyses at endosomal pH 5.5, shedding the PEG corona and exposing the DOPE/CHEMS fusogenic lipid layer for mitochondrial docking.

4.2 Layer 2 -- Cardiomyocyte Selectivity and Endosomal Escape

- **AT1R targeting peptide** provides cardiomyocyte vs. immune cell discrimination. AT1R is highly expressed on cardiomyocytes and low/absent on neutrophils and macrophages. Included in the full design (Phase 2); omitted from MVD (Phase 1).
- **DOPE/CHEMS fusogenic lipid coat** (7:3 molar ratio). At pH 7.4: lamellar phase, stable. At pH 5.5 (endosome): hexagonal phase transition, membrane fusion, endosomal escape. Escape efficiency 5-15% vs ~1-3% for unmodified NPs (Allen & Cullis 2013).

- **SS-31 (elamipretide sequence)** -- D-Arg-Dmt-Lys-Phe-NH2 -- conjugated to DOPE headgroups after PEG shedding. Binds cardiolipin on the outer mitochondrial membrane via electrostatic and hydrophobic interactions. Targeting is ΔΨm-independent. FDA accelerated approval of elamipretide for Barth syndrome (Forzinity, September 2025) formally validates this CL-binding mechanism.

4.3 Layer 3 -- AND Gate Release Mechanism

The AND gate requires co-activation of two independent oxidation-sensitive polymer arms to trigger shell disassembly. The shell is a block copolymer co-assembly: Polymer A (dioxaborolane pendant, Arm A) and Polymer B (O-alkylhydroxylamine pendant, Arm B) co-precipitated by nanoprecipitation. Both must reach oxidation threshold; either alone is insufficient.

Arm A: Dioxaborolane -- H2O2 Temporal Integrator

Dioxaborolane boronate esters react with H_2O_2 in a pH-dependent manner. At matrix pH 8.0, pH-corrected $k_2 = 600\ M^{-}s^{-}$ (Bhattacharya & Chang JACS 2014). The boronate ester is completely GSH-inert. Slow kinetics acts as a temporal low-pass filter: transient H_2O_2 from exercise does not accumulate sufficient integral to cross threshold, while sustained dox-induced H_2O_2 does. Arm A fires at t = 22 min in the ODE model trajectory.

Arm B: O-Alkylhydroxylamine Oxime -- 4-HNE Lipid Peroxidation Marker

Arm B is a three-branch contingent design gated by Experiment 0. This experiment must precede polymer synthesis; its result determines which linker chemistry is implemented.

Branch	[4-HNE] Criterion	Linker	k2 (M-1s-1)	t_gate (min)	Protection
A	>= 0.075 uM	Methoxyamine	300	~120	83-99%
B	0.025-0.075 uM	Benzyloxyamine	1,000	~49-108	83-87%
C	< 0.025 uM	Benzyloxyamine + MDA	50 (MDA)	~51	~87%

4.4 Layer 4 -- Payload: MnTE-2-PyP

MnTE-2-PyP (manganese(III) meso-tetrakis(N-ethylpyridinium-2-yl)porphyrin, MW 900 Da) is the primary payload. Selection rationale:

- **kcat/Km for O2- = 7x10^{8} M^{-}s^{-}** (Batinic-Haberle 2009, 2012). At achievable matrix concentrations, k_sc_O2 = 23,515-141,000 min-1 -- over 4,700x above the k_sod threshold of 5 min-1.
- **No Fenton risk** -- Mn(III) porphyrin reduction potential (+0.228 V vs NHE) is thermodynamically unfavourable with GSH. Unlike Fe-porphyrins or Mn-salen complexes.
- **No phosphate competition** -- porphyrin macrocyclic ligation is too strong for matrix phosphate (10-20 mM) to displace Mn, unlike salen complexes where phosphate reduces activity 3-5x (Kimura 2007).
- **DeltaPsim-independent accumulation** -- MW 900 Da, crosses OMM via VDAC passively. Retains full activity in partially depolarised dox-stressed mitochondria where MitoQ fails.
- **Phase II clinical data** -- Duke group (Batinic-Haberle/Hadley) has demonstrated in vivo efficacy and safety in multiple Phase II trials.

Mn(II) cyclen was evaluated and eliminated. Matrix corrections (pH, 20 mM phosphate competition, GSH-mediated redox inactivation) reduce effective kcat/Km to ~1.3x10^{4} M^{-}s^{-}, delivering k_sc = 1.5 min-1 -- 3x below requirement. MnTE-2-PyP strictly dominates.

5. Payload Encapsulation: Contingent Design (Experiment 1A)

MnTE-2-PyP carries net +5 charge at physiological pH and logP approximately -3 to -5 -- incompatible with standard nanoprecipitation (EE ~1-5%) and in conflict with double emulsion methods (dioxaborolane boronate esters hydrolyse in the aqueous outer phase required for w/o/w emulsion). Three encapsulation methods are attempted sequentially.

- **Method 1 -- Modified nanoprecipitation with DMSO cosolvent:** DMSO (20-30% v/v) disrupts hydrophilic/hydrophobic partitioning. Target EE: 5-15%. Success criteria: EE > 5% AND boronate retention > 85%.

- **Method 2 -- Electrostatic complexation with polyanionic block:** Poly(acrylic acid) block introduced as third co-assembly polymer. Ion pairing between MnTE-2-PyP (+5) and PAA carboxylates. Literature EE: 30-70%. AND gate function preserved.
- **Method 3 -- Sequential surface loading:** MnTE-2-PyP electrostatically adsorbed to pre-formed NP surface. Critical consequence: the AND gate no longer gates payload release -- converts to constitutive delivery. Still provides ceiling-level cardioprotection from moment of mito contact.
- **If all three fail:** Design bifurcates into two co-administered NP populations. Population 1: dual-polymer AND gate NPs (no payload). Population 2: standard PEG-PLGA NPs encapsulating MnTE-2-PyP at 5% EE with constitutive PLGA-degradation release. AND gate function is lost from payload release but retained as a diagnostic readout.

6. Co-Assembly Architecture: Contingent Design (Experiment 1B)

AND gate function requires that each individual NP contains both Polymer A and Polymer B chains. Phase separation would produce single-trigger particles. Experiment 1B characterises co-assembly homogeneity by FRET using donor-labelled Polymer A (coumarin) and acceptor-labelled Polymer B (rhodamine). Decision threshold: FRET efficiency > 5% confirms homogeneous mixing.

- **Outcome 1 -- FRET confirms mixing (efficiency > 5%):** No compatibiliser required. Proceed to AND gate functional validation.
- **Outcome 2 -- FRET indicates phase separation (< 1% or bimodal):** Compatibilising Polymer C added at 10-20 wt%. Since Polymer C contains both pendant types, chi_AC ~ 0 and chi_BC ~ 0, driving the system into the homogeneous mixing regime.
- **Outcome 3 -- Compatibiliser fails:** Sequential addition nanoprecipitation creates a layered structure (Polymer A core, Polymer B shell). H_2O_2 diffusion through the 20 nm Polymer B shell requires tau ~ 20 microseconds, six orders of magnitude faster than boronate oxidation timescale. AND gate is preserved in both geometry and function.

7. Delivery Cascade: Quantitative Summary

At the recommended clinical scenario (IV administration, 20 mg/kg, CRPPR + ICAM-1 dual targeting):

Step	Efficiency	NPs Remaining	Basis
Injected	100%	1.1 x 10^15	20 mg/kg, 100 nm NP
Cardiac accumulation	~2%	2.2 x 10^13	CRPPR + ICAM-1 dual targeting
Cardiomyocyte uptake (AT1R)	~60%	1.3 x 10^13	Receptor-mediated endocytosis
Endosomal escape (DOPE/CHEMS)	~10%	1.3 x 10^12	pH-triggered membrane fusion
Per cardiomyocyte (cytoplasm)	--	~357 NPs/cell	7.5 x 10^9 cardiomyocytes
Payload delivered (EE = 5%)	--	~2.9 x 10^9 molecules/cell	MnTE-2-PyP at 5% EE
Matrix k_sc_O2 achieved	--	~23,500 min-1	vs k_sod = 5 min-1
Safety margin	--	~4,700x	above k_sc floor

> Even with 3% endosomal escape (MVD, without DOPE/CHEMS) and 1% cardiac accumulation: 88 NPs/cell cytoplasm, k_sc_O2 approximately 5,800 min-1 -- still 1,160x above requirement. The design carries sufficient margin that endosomal escape efficiency and cardiac accumulation efficiency are not bottlenecks to efficacy within their realistic ranges.

8. Manufacturing and Minimum Viable Design

8.1 The Manufacturing Constraint

The full 12-component design has an estimated synthesis probability of 7.5% on first attempt and batch-to-batch variability of +/-75% CV. The most complex approved nanomedicine (Onpattro) has 5 components and required 23 years from concept to approval.

The load-bearing synthesis incompatibility: dioxaborolane and O-alkylhydroxylamine cannot be polymerised in the same pot -- hydroxylamine reduces boronate esters during polymerisation. Resolution: two-polymer co-assembly synthesised separately and co-assembled by nanoprecipitation.

8.2 The Minimum Viable Design (MVD)

The MVD retains all five novelty claims and the complete AND gate mechanism while reducing synthesis complexity to a level achievable in a materials chemistry research lab.

Component	Full Design	MVD	What Is Lost
Polymer A (boronate)	Yes	Yes	Nothing
Polymer B (oxime)	Yes	Yes	Nothing
Mn payload	Cyclen + MnTE-2-PyP	MnTE-2-PyP only	Minor upstream redundancy
Cardiac targeting	CRPPR + ICAM-1 + AT1R	CRPPR + ICAM-1 + CD47	Cardiomyocyte selectivity (AT1R)
Endosomal escape	DOPE/CHEMS (10%)	DOPE/CHEMS (10%)	Nothing -- retained
pH-labile PEG	Hydrazone linker	Hydrazone linker	Nothing -- retained
Mito docking	SS-31 conjugated	Omitted (Phase 2)	OMM proximity
Components	~12	~5	--
Synthesis probability	7.5%	~75%	--
Batch-to-batch CV	+/-75%	+/-41%	--

9. Patient Stratification and Clinical Translation

9.1 Target Population

The design targets adult patients with breast cancer or lymphoma receiving doxorubicin-containing regimens classified as moderate to high baseline cardiovascular risk using the ESC 2022 Cardio-Oncology Risk Stratification Framework (Lyon et al. Eur Heart J 2022): specifically those with 2 or more ESC 2022 risk factors or planned cumulative doxorubicin dose 250 mg/m^2 or above.

9.2 Biomarker-Guided Co-Administration

NP co-administration begins from cycle 3 onward in patients whose hs-cTnT exceeds 14 ng/L or whose NT-proBNP increases more than 30% above baseline during cycles 1-2. This decision rule aligns with the NP's self-amplifying accumulation mechanism: the biomarker rise that triggers NP co-administration is the same signal driving ICAM-1 upregulation that enhances NP accumulation.

9.3 Trial Design

Parameter	Specification
Design	Phase II randomised controlled trial
Intervention	Standard of care + dexrazoxane + NanoROS vs SoC + dexrazoxane
Primary endpoint	hs-cTnT at cycle 6 relative to cycle 1 (subclinical damage)

Parameter	Specification
Secondary endpoint	LVEF at 6 months post-chemotherapy completion
Exploratory endpoint	Cardiac MRI at 12 months (fibrosis, strain imaging)
Sample size	120-180 patients (80% power, 20% additional reduction above dex)
Biomarker trigger	hs-cTnT > 14 ng/L or NT-proBNP > 30% rise during cycles 1-2

10. Experimental Roadmap

All experiments are ordered by dependency. Earlier experiments gate later ones. Experiments 0 and 1A are pre-synthesis gates -- they must be completed before any polymer synthesis work begins.

Exp.	Name	Go Criterion	If Fail
0 *	Free [4-HNE] LC-MS/MS (dox-treated CM mito fraction)	[4-HNE]_free detectable >= 10 nM	Branch C: switch to MDA sensing
1A *	MnTE-2-PyP encapsulation feasibility (Methods 1-2-3)	EE > 5% AND boronate retention > 85%	Alternative A: separate NP populations
1B	FRET co-assembly homogeneity (Polymer A + B)	FRET efficiency > 5%	Compatibiliser C or sequential addition
2	pH-responsive PEG shedding kinetics	> 80% PEG shed at pH 5.5 in 2 hr	Switch to acetal linker
3	DOPE/CHEMS endosomal escape (neonatal rat CMs)	> 5% calcein delivery vs macrophage control	Add KALA peptide
4	AND gate specificity (healthy vs dox-treated CMs)	< 5% release healthy; > 50% dox-treated	Increase oxime threshold
5	CL-OOH and [4-HNE] kinetics fitting (isolated mito)	k_CL_ox and k_HNE_prod fitted to HPLC-MS data	Accept parametric uncertainty
6	Cardiomyocyte vs macrophage selectivity	> 10x CM enrichment vs cardiac macrophage	Optimise AT1R peptide density
7	In vivo cardiac biodistribution (mouse dox model)	>= 1% ID/g cardiac tissue	Add ultrasound-mediated delivery
8	Functional cardioprotection (6-cycle mouse dox model)	>= 20% reduction cardiac troponin at cycle 3	Increase dose or intra-coronary route

> * = Pre-synthesis gate. Experiments 0 and 1A must be completed before any polymer synthesis begins. Their results determine linker chemistry selection and payload encapsulation strategy respectively.

11. Limitations and Honest Failure Mode Landscape

11.1 Unanchored Quantitative Dependencies

- **Free [4-HNE] in cardiomyocyte mitochondria under dox exposure has not been directly measured.** Estimated range 0.034-0.46 uM spans more than an order of magnitude. Linker chemistry selection cannot be finalised without Experiment 0.
- **MnTE-2-PyP encapsulation in dioxaborolane polymer NP is unconfirmed.** The boronate hydrolysis conflict under aqueous conditions has no published resolution. Encapsulation feasibility is load-bearing for the payload chapter.

- **k_CL_ox is estimated, not measured.** Current estimate calibrated to literature [CL-OOH]_ss = 5 nmol/mg but not measured in dox-treated cardiomyocytes specifically.

11.2 Design Failures That Require Bench Work

If [4-HNE] < 0.025 uM (Branch C outcome), AND gate Arm B is redesigned to sense MDA -- a chemistry with less published validation for NP applications and a bifunctionality aggregation risk.

If all three MnTE-2-PyP encapsulation methods fail, the design bifurcates into separate NP populations and the chemical AND gate is lost from the payload release mechanism. Specificity then depends entirely on the targeting cascade.

Phase separation in Polymer A/B co-assembly would produce a mixed population of single-trigger NPs. Without the FRET experiment, this failure is silent.

11.3 Human Biodistribution Uncertainty

CRPPR cardiac accumulation was demonstrated at 4.2% in mice (Dvir 2011). Human cardiac accumulation is estimated at 0.3-1.0%. If human accumulation is below 0.5%, dose must be increased or intra-coronary administration route used (30% cardiac accumulation at 5 mg/kg, requiring catheterisation).

11.4 The TAZPOWER Timeline Signal

Elamipretide's clinical benefit in Barth syndrome required 168 weeks of open-label extension to manifest as functional improvement. Phase 1 mouse model experiments should evaluate cardiac function endpoints at cycles 1, 3, and 6 to establish the earliest timepoint of detectable functional benefit.

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